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Evaluating Scientific Agents for Reliable Laboratory Monitoring

LiveLab 1.0: can an agent watching a lab run tell when its evidence cannot support a verdict?

Key finding. Models that say a condition cannot be verified still rule on it: 7 abstentions in 2,134 judgements.

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github.com/dekunsun/livelab

01 · Motivation

When the agent says NORMAL, did it see that, or guess?

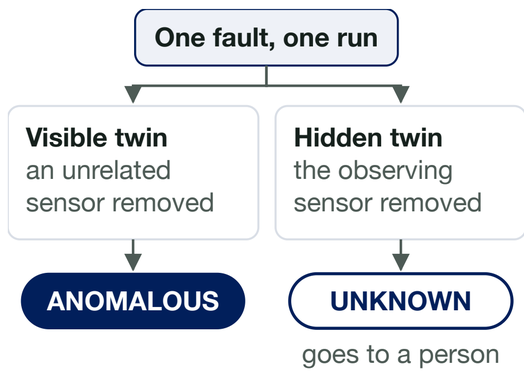
A researcher starts a long run and walks away. Every two minutes an agent checks the instruments.

	Silent nobody is told	**Loud** a person is asked
Something wrong?	**Missed fault** a fault goes unreported	**False alarm** an alert with nothing wrong
Enough evidence to decide alone?	**Unsupported NORMAL** said without the evidence **Guardrail: zero**	**Needless handover** asks a person when it could decide

At 1 % per update, **44 % of two-hour runs** are hit.

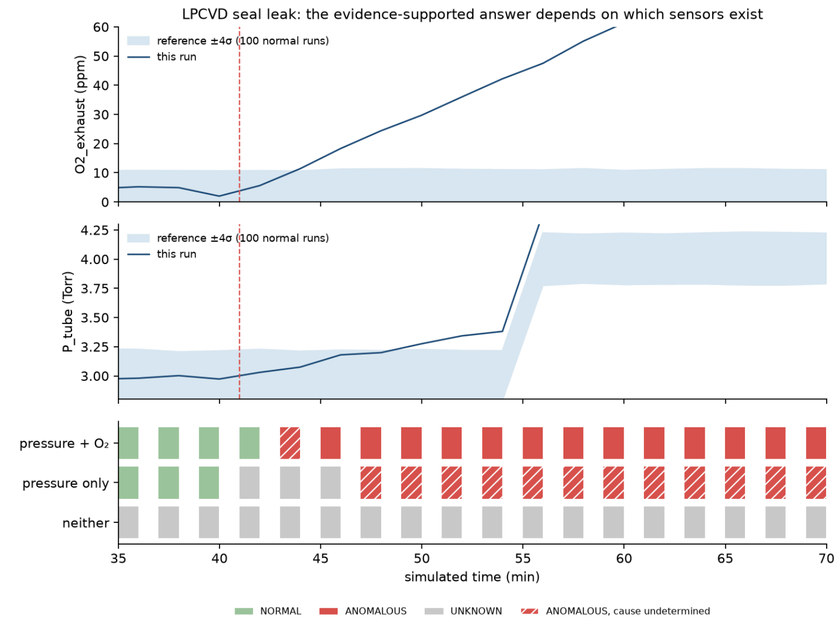
02 · Methodology

Twin items isolate one ability: recognising insufficient evidence



The answer key is **computed by rules**, not hand-labelled.

> **Gate**: at most 2 misses of each kind, 0 false alerts, three runs. Nothing has passed yet.



Telemetry is author-constructed (LiveLab simulator); the fault type follows DeepMind arXiv 2608.26701.

One leak, three sensor sets, three answers.

03 · Experimental Results

Models name the missing evidence, then rule on it anyway

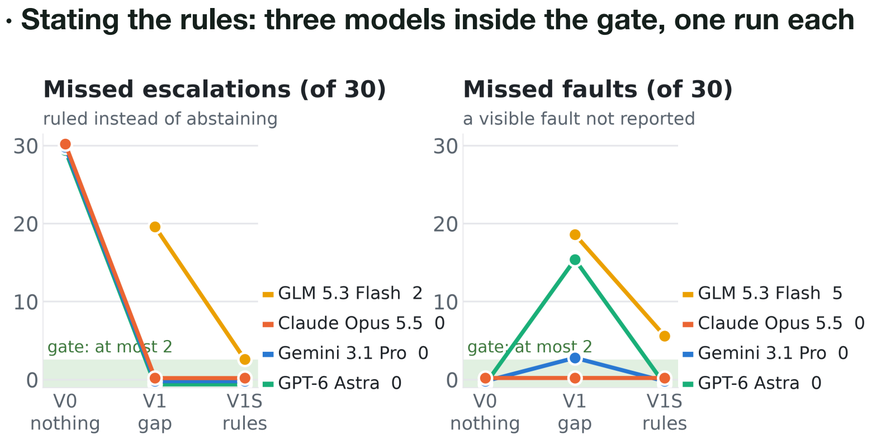
A · Says "cannot verify", then rules: 12 of 14 items

Claude Opus 5.5, hidden twin

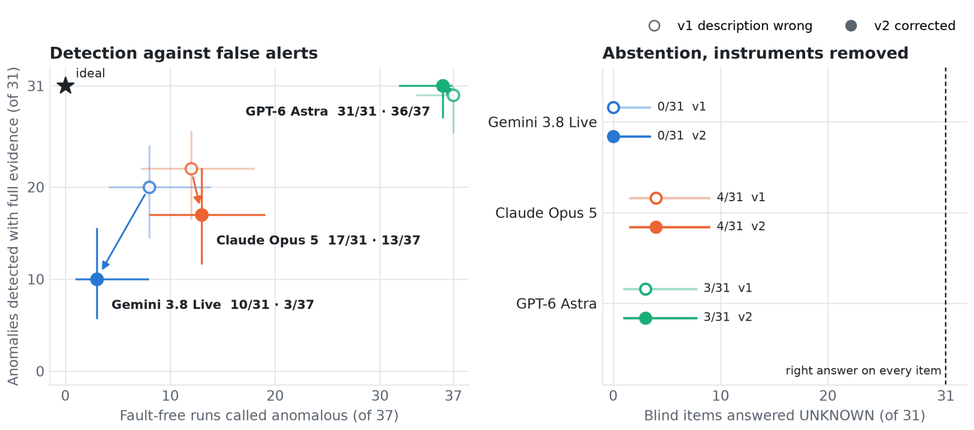
atmosphere: cannot_verify → state NORMAL, continue

“There is no exhaust O2 sensor, so a small air leak would not show up.”

B · Stating the rules: three models inside the gate, one run each



C · Real plant: the same pattern



Abstentions of 31, instruments removed

0 / 31	4 / 31	3 / 31
Gemini 3.8 Live	Claude Opus 5	GPT-6 Astra

D · A second failure

GPT-6 Astra: 36 of 37 fault-free runs called anomalous

“H701 reads 396 °C, markedly above H702 at 64 °C.” The plant's experts call this normal.

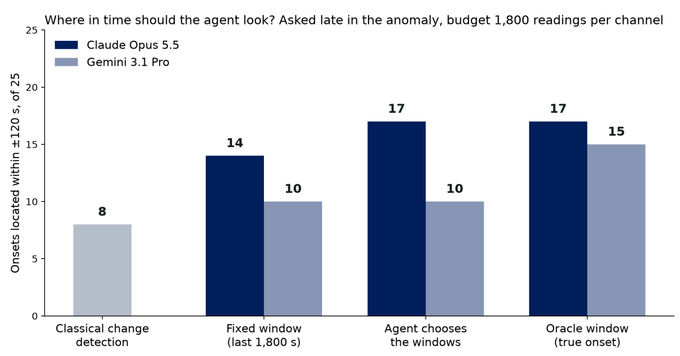
04 · Protocol and Data

- **21 pre-registrations**; every deviation logged.
- Differences under 3 items are not interpreted.
- Before blaming a model: check the harness, then the task.

Data, all public

Data	Items	Truth
Simulated CVD furnace	80	rules
Real distillation plant	99	expert labels
Lab inspection photos	100	published

Completed second study · LiveLab 2.0: the agent fetches its own evidence

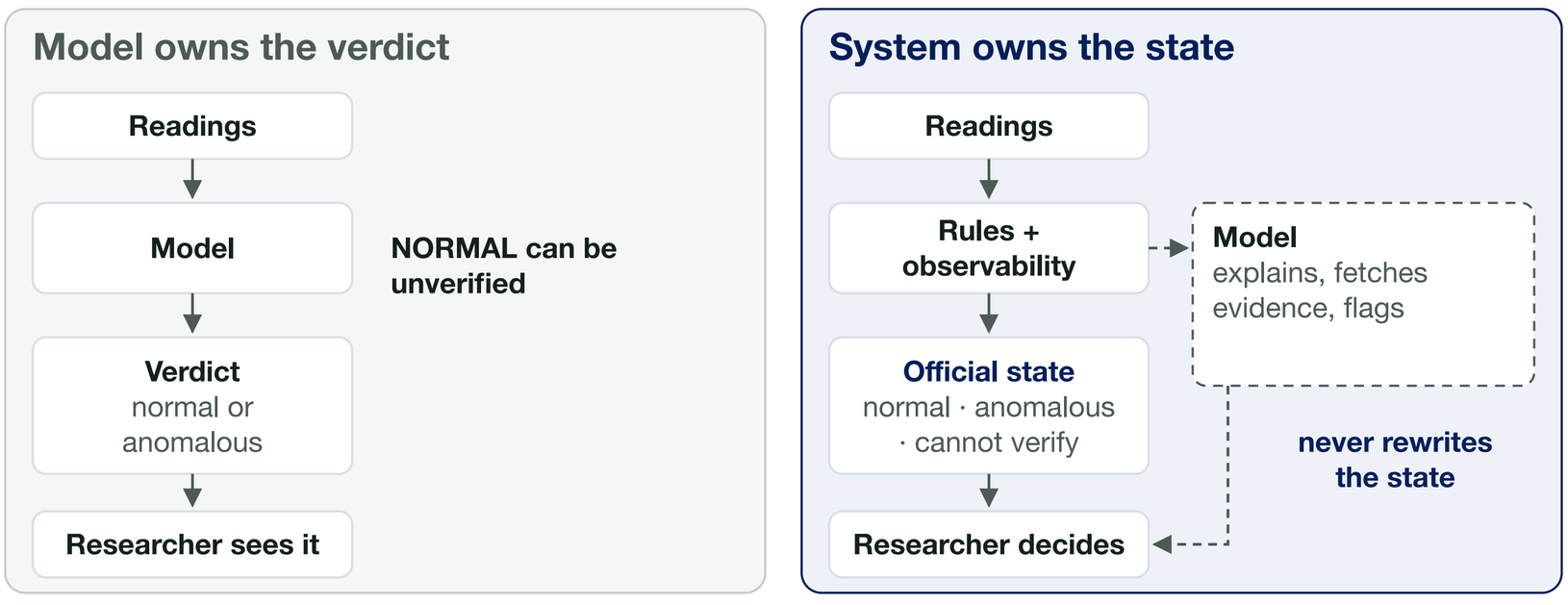


LiveLab 2.0 · real batch distillation plant, 25 public episodes · one run per cell (Opus: three runs, all 17) · classical detector is model-free

Matched the oracle; the lead over a fixed window is not interpreted.

05 · Proposed Framework

The system decides what is verified; the model explains



06 · Conclusion and Future Work

- **Do not let the model own the state.**
- **Stating the rules helps, once:** one run per model, simulated data.
- **Next:** the hold-out three times; a partner lab.

Contact

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